

Free-weight augmentation with elastic bands improves bench-press kinematics in professional rugby players

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Short title: Kinematics of bench press with elastic resistance superimposed

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ABSTRACT

This study aimed to investigate the effects of combining elastic bands to free weight (EB+FWR) on the acceleration-deceleration and velocity profiles of the bench press in professional rugby players and recreationally-trained subjects. Sixteen male subjects (8 rugby players and 8 recreationally-trained subjects) were randomly assigned to complete 2 experimental conditions in a cross-over fashion: EB+FWR and FWR. In both conditions, subjects performed one bench-press set to volitional exhaustion with a load equivalent to the 85% of one-repetition maximum (1RM). In the EB+FWR condition, the contribution of elastic resistance was approximately the 20% of selected load (85% of 1RM). Results indicate that EB+FWR condition increased significantly the range of concentric movement in which the barbell is accelerated. This increase was significantly higher in rugby players (35%) in comparison with recreationally-trained subjects (13%). Maximal velocity was also increased in EB+FWR (17%), when compared with FWR condition. These results suggest that when combined with variable resistance (i.e. EB), the external resistance seems to be more evenly distributed over the full range of motion, decreasing the need for dramatic deceleration at the end of the concentric phase. The present data also indicate that the kinematic benefits of a EB+FWR approach seems to be more prominent in athletes from modalities in which high level of strength and power are required (i.e. rugby players).

Keywords: elastic bands; kinematics; bench-press; rugby players

INTRODUCTION

Optimal combinations of muscle strength and power are critical to maximize athletic performance in several sport modalities. In this sense, kinematics associated to resistance exercises (e.g. velocity and acceleration) have been proposed as one of the most important stimuli for maximal strength and muscle power resistance training-induced adaptations (12). Many studies have investigated kinematics associated to different resistance exercises and loads performed by different training-level subjects

(3,6,8,14). Thus, different strategies have been proposed in order to optimize the kinematics of resistance exercise (i.e. ballistic training and variable-resistance training). Variable resistance is designed to change the external resistance load throughout an exercise's range of motion (i.e. elastic bands, bungy bands and/or chains attached to the ends of bars) (13). In this line, in an attempt to merge the benefits of variable resistance with those of constant external loads, combinations of elastic bands (EB) with free weight resistances (FWR) have increased in popularity (1,2,4,7,10,15,16,19). Although a recent study analyzed the effect of EB applied "in series" to FWR (7), the EB are usually applied in a "in parallel" approach; that is, the EB are attached to the end of the barbell so that as the barbell ascends during the concentric phase (i.e. during a back squat or a bench press) an increasing load is assumed as the EB lengthens. One of the potential benefits of adding EB to free weights is the reduction of the large deceleration period that occurs at the end of concentric motion (19). The deceleration phase is an important part of the lift, in which bar velocity decreases because it is unintentionally decelerated by the performer (14). Thus, there have been reported accelerative portions ranging from 60% to 66% in bench press for loads equivalent to 45% to 60% of 1RM (6,8,14). However, to the best of our knowledge, the acceleration-deceleration profile on an EB+FWR approach has not been analyzed. Moreover, studies focused on EB+FWR effects over kinematics usually analyze recreationally trained (10, 16,19) or untrained subjects (2). Thus, to our knowledge, only three studies (1,4,9) have analyzed the effects of elastic resistance superimposed to free weight in competitive athletes. Therefore, the purpose of the present research was to study the effects of combining elastic resistance to free weight (EB+FWR) on the acceleration-deceleration and velocity profiles of the bench press in professional rugby players and recreationally trained subjects. We hypothesized that this combination should decrease the need for dramatic deceleration during the movement and, as a result, an enhanced average muscle tension could be achieved throughout the concentric range of motion. Moreover, the training level (professional rugby players vs. recreationally trained subjects) could also play a significant role over the effects of EB+FWR training approach.

METHODS

Experimental Approach to the Problem

A randomized cross-over study was designed to compare bench-press performance during a set to failure with (EB+FWR) and without (FRW) elastic resistance

superimposed in subjects with a different training status profile: professional rugby players vs. recreationally resistance-trained subjects (controls). Data collection took place over a period of four weeks with one experimental session each week. Eight professional rugby players and eight healthy undergraduate students participated in the study. The first two sessions were used to familiarize subjects with testing procedures and to assess the subjects' one-repetition maximum (1RM) in the bench press. During each of the next two testing sessions, one set of the bench press was carried out, leading to volitional exhaustion, with a load equivalent to 85% of 1RM. During each one of such testing sessions, one of the two conditions was performed: EB+FWR and FWR. For the EB+FWR condition, the proportional contribution to total resistance consisted of 21% EB and 79% standard free weight. The average force produced by the bands was estimated from regression equations provided previously (15). For the FWR condition only standard weight was used. The sequence of testing sessions was counterbalanced to control for potential order effects.

Subjects

Sixteen male subjects volunteered for the study: eight undergraduate students and eight professional rugby players. All of them reported high experience with free-weight resistance exercises and training leading to failure. At the time of the study and from one month before, none of the subjects were engaged in any regular or organized resistance training programme. None of them reported using anabolic steroids. The biometric data and characteristics of the sixteen subjects are presented in Table 1.

The rugby players were members of the same team (*Club de Rugby Cetransa el Salvador*), which is playing in the *División de Honor*, which is the top competition in Spain. Concerning the annual periodization of the rugby team, the study was carried out during the beginning of the transition phase (April to June), one month after the season's end. This phase is used to facilitate psychological rest, relaxation and biological regeneration.

The undergraduate students were physically active and they had not been involved in a strength-training regimen within the previous month, although all of them averaged at least 6 months' experience with free-weight resistance exercises. Their normal workouts

typically lasted just less than 90 min and entailed training of multiple body parts and exercises.

Prior to data collection, subjects were informed of the requirements associated with participation and provided written informed consent. Moreover, subjects did not allow their sleeping, eating, and drinking habits to change throughout study participation. The research project was conducted according to the Declaration of Helsinki and it was approved by the University Review Board for use of Human Subjects.

Procedures

Data collection took place over a period of 4 weeks with 1 testing session each week. In the first session, instructions regarding preparation for the one-repetition maximum (1RM) testing and proper form were given to each participant. Moreover, subjects were familiarized with the EB+FWR condition. During the second experimental session, the assessment of the 1RM for the bench press was determined. During each of the next 2 testing sessions, 1 set of the bench press was performed to failure. During such testing sessions, one of the two conditions designed was performed: EB+FWR and FWR condition. A counterbalance procedure was used to determine the condition for each testing session. Thus, at the end of the experimental phase, all the subjects had been tested for the two conditions. Testing sessions were carried out the same day of the week, in all cases at the same time of the day.

Maximal strength measurement and elastic-bands settings

1RM bench press was assessed using a previously established protocol (17). Briefly, after a light warm up on the bench press, subjects attempted to lift a progressively increasing load, allowing 3 minutes of resting periods between attempts. 1RM value was obtained using as few attempts as possible (5 attempts as maximum). Regarding bench press technique, subjects were asked to keep their head, shoulders/upper-back area, and buttocks in contact with the bench and both feet securely on the floor during the whole movement. Hence, for the eccentric phase, they had to lower the barbell until the chest was touched lightly approximately 3 cm superior to the xiphoid process, avoiding to bounce the barbell off the chest or to lift the buttocks off the bench when raising the barbell. Then, they were encouraged to push the barbell up and slightly

backward forcefully, until elbows were fully extended. Back arching was prevented during the whole repetition. Hand spacing was adjusted individually to no more than 1.5 biacromial width, recording this distance for each individual in order to maintain grip width throughout the experimental sessions.

Two elastic bands (Flex Bands, elite FTS; London, Ohio) of varying thickness (orange and red) were used in the subsequent EB+FWR condition, in a draped bench press configuration, as described previously by Shoepe and coworkers (15). Briefly, the band was wrapped under the cross-support with no knot while both free ends were affixed to the barbell producing 4 lengths of elastic resistance from the same band (Figure 1). This configuration was necessary because of the length of the bands and the inability to tie either end to a ground support on this equipment.

Given that contribution of EB varies through position, this first experimental session was used to measure various lengths of each-subject range of motion. Thus, the average force produced by the bands was estimated from regression equations provided by Shoepe and coworkers (15), whose R^2 values exceeded 0.9882. Because of similarities in subjects, only the red and the orange bands were needed during the protocol, to ensure the elastic-resistance required. An elastic contribution of approximately the 20% of selected load (85% of 1RM) was chosen as this proportion has been used previously in different studies (1,16,19). Once the regression equations were applied, the average contribution of elastic resistance in EB+FWR condition was 21.71 ± 4.45 % of total load (85% of 1RM)

Bench press sets to failure

Each bench-press protocol consisted of performing one set to volitional exhaustion, with a load equivalent to subject's 85% of 1RM. In EB+FWR and FWR conditions subjects began with a warm-up consisting of 5 minutes of low-resistance lower-body cycling on an ergometer followed by two bench-press sets. The first warm-up set consisted of ten repetitions at 30% 1RM and the second warm-up set consisted of ten repetitions at 50% 1RM, allowing 1 min of resting between sets. Two minutes after the specific warm up, individuals began the bench-press set. Thus, they were asked to move the barbell as fast as possible during the concentric phase of each repetition, until

volitional exhaustion. Failure was defined, according to a previously established criterion (11), as the time point when the barbell ceased to move, if the subject paused more than 1s when the arms were in the extended position, or if the subject was unable to reach the full extension position of the elbows. Subjects were asked to maintain a correct bench press technique, as described previously. Kinematic parameters of each repetition were monitored by linking a rotary encoder (Globus Real Power, Globus, Codogne, Italy) to the end of the barbell. The rotary encoder recorded the position of the barbell within an accuracy of 0.1 mm and time events with an accuracy of 0.001 s. Total repetitions, as well as mean velocity (considering whole set), maximal accelerative portion and mean accelerative portion (considering the whole set) were analyzed. The test-retest intraclass correlations coefficients for all dependent variables were greater than 0.78 and the coefficients of variation (CV) ranged from 0.9% to 2.2%.

Statistical Analyses

Normality of the dependent variables was checked and subsequently confirmed using the Kolmogorov–Smirnov test. Differences concerning number of repetitions achieved, maximal velocity, mean velocity (considering whole set), maximal accelerative portion and mean accelerative portion (considering the whole set) were analyzed through a two-way ANOVA. The 2 factors were condition (FWR vs. EB+FWR) and group (rugby players vs. controls). When a significant F-value was achieved, pairwise comparisons were performed using a Bonferroni *post hoc* procedure. Statistical significance was set at $p < 0.05$.

RESULTS

Descriptive data

Subject characteristics are listed in Table 1. Rugby players were heavier (35.6%; $p < 0.001$), older (29.4%; $p < 0.01$) and stronger in the bench press 1RM (81.8%; $p < 0.001$) than their recreationally-trained counterparts.

Number of repetitions

Although repetitions performed in EB+FWR were slightly higher than FWR condition (table 2), no significant group, condition nor group x condition effects were observed ($p > 0.05$).

Accelerative portion of the concentric phase

Significant condition ($p<0.01$) and group x condition ($p<0.001$) effects were observed (Fig. 2) regarding the maximal proportion of concentric movement time in which barbell was accelerated (AP_{max}). The repetition with the AP_{max} corresponded to the first or the second repetition. Both groups enhanced significantly the AP_{max} in EB+FWR when compared to FWR (35 and 13% for rugby players and controls respectively), although the increase in rugby players was significantly higher than controls ($p<0.05$).

Table 2 displays the average accelerative portion for the whole set. Although slight increases in this variable were observed when the EB+FWR condition was compared to the FWR condition (8% approximately), no group, condition nor group x condition significant effects were pointed out by statistical analysis.

Velocity of the concentric phase

Figure 3 shows maximal velocity (V_{max}) obtained throughout the set for both conditions in rugby players and controls. V_{max} was achieved within the first two repetitions. A significant condition effect ($p<0.05$) was observed. That is, the use of elastic bands allowed an increase in V_{max} (~17%), with no differences between groups.

Mean velocity of the whole set (V_{mean}) is shown in Table 2. Again, a significant condition effect was observed ($p<0.05$). However, post hoc analysis pointed out that the increase obtained when using elastic bands (14.5 and 6.9% for rugby players and controls, respectively) was only statistically significant for rugby players ($p<0.05$).

DISCUSSION

The primary finding of the present study is that elastic resistance applied “in parallel” to conventional bench press (EB+FWR) increases the range of concentric movement in which the barbell is accelerated (AP_{max}), as well as maximal velocity (V_{max}). Traditional FWR exercises are characterized by an initial peak force phase after which a dramatic deceleration phase takes place during the later stages of the concentric contraction (3,6,14). Ballistic exercises (i.e. bench throws and squat jumps) have been prescribed to solve the deceleration problem in conventional FWR (14). However, ballistic training is not easily applicable to all the exercises commonly performed in resistance training

facilities, in order to improve the kinematics associated with FWR. A different possibility proposed to improve the kinematics associated to FWR exercises is the combination of EB with FWR (1,4,19). In combination with EB, the external resistance during an exercise should theoretically be more evenly distributed over the full range of motion than with FWR exercise. Therefore, this combination should decrease the need for dramatic deceleration during the movement and, as a result, an enhanced average muscle tension could be achieved throughout the range of motion, as it has been hypothesized previously (1). To our knowledge, this is the first study demonstrating the increase in the bench-press AP_{max} when an EB+FWR approach is applied. This positive effect of combining resistances of two different nature seems to be larger when subjects are highly trained (i.e. professional rugby players). The mechanism accounting for this between-group difference is unclear, but an alteration in recruitment patterns is a conceivable explanation. Contraction with EB+FRW should have reduced the biomechanical disadvantages throughout the concentric range of motion, which may have kept the muscle working closer to maximal capacity. In fact, contractions with a less acute sticking point may have elicited greater type IIx muscle fiber recruitment, as suggested by Anderson and coworkers (1). Given the resistance-training nature usually employed in top-level rugby, it makes sense to suppose that rugby players are more used to maximal strength and power efforts, in comparison with recreationally trained subjects. Therefore, it could be speculated that highly-trained subjects may get a large benefit when using an EB+FWR training approach.

During the FWR condition, the AP_{max} observed in the present research is lower than that reported in previous studies analyzing the bench press with loads equivalent to 45% of 1RM (13) or 60% of 1RM (5,7) in different populations. Investigating a power-lifters sample and using similar loads (81% of 1RM), Elliot and coworkers (5) found the AP_{max} to be 48.3% of the concentric movement time. Some methodological differences existing between that study and the present research could explain the lower AP_{max} observed by Elliot and coworkers, even though they used a lower load. They allowed only one attempt, and previous studies indicate that first repetition is used to gradually achieve the maximum repetition velocity (11), which agree with our results. In fact, studies focused on peak power measurement during free weight movements usually elicit 3 repetitions (18). When using an EB+FWR approach, rugby players showed an AP_{max} equivalent to 81% of concentric movement, which resulted in a significantly

higher $V_{\max}$. This $AP_{\max}$ is near to that observed previously in ballistic bench throws with much less intensity (14).

Performing sets of multiple repetitions is the inherent nature of a typical strength-training session. In this sense, the present research used a repetitions-to-failure approach. Although rugby players experienced an increased mean velocity considering the whole set, the kinematics of a set to failure seems to be less affected by an EB+FWR training approach. It could be elucidated that the fatigue induced by a set carried out until volitional exhaustion may reduce the mechanical advantage of using elastic resistance superimposed to free weight. Future experimental designs should test this hypothesis using different absolute intensities (% 1RM) as well different relative resistances provided by EB and FWR, respectively.

One of the limitations that coaches and athletes can find when using elastic resistance is the quantification of the external load applied, in order to normalize the EB+FWR condition to the de FWR condition. The most common way to calculate the amount of resistance provided by EB is to set them up so that the EB+FWR, after the lifter has achieved full extension, is equal to the resistance that would otherwise come from isolate FWR (19). Given that EB increases tension progressively throughout the concentric phase of the bench press, this method does not properly normalize the external load. In the present research, the load added by EB has been estimated as a mean value, considering two points throughout each-subject range of movement. Although this method does not solve completely the problem of normalization of external load, at least it is minimized. The fact that the number of repetitions achieved was similar in both experimental conditions invites to speculate that the external load throughout the whole range of motion was similar.

In summary, performing the bench press with a load of 85% 1RM composed of free weight resistance (80%) and elastic resistance (20%) increases the range of concentric movement in which the barbell is accelerated, as well as maximal and whole-set mean velocity, in professional rugby players. These positive effects have been also observed in recreationally trained subjects, although in a lesser extent. The maximal strength and power contractions included in common resistance training routines for elite rugby players could make them especially susceptible to mechanical advantages or elastic

resistance superimposed to traditional free weight. Future studies would be needed to confirm this speculation.

PRACTICAL APPLICATIONS

It is generally accepted that the more specific a training exercise to a competitive movement, the greater the transfer of the training effect to performance. Although free weight resistance (FWR) is the most popular mode of resistance training, a substantial portion of the lifts involve a period when the load is decelerated. Variable resistance (i.e. elastic bands; EB) has lately gained much attention in this sense, since it may change the external resistance load throughout an exercise's range of motion. The findings of this investigation demonstrate that applying EB in parallel with FWR may enhance some kinematic parameters of a high-intensity bench press set. These positive effects seem to be more prominent in athletes whose modalities require high levels of strength and (i.e. rugby players). Thus, strength-based athletes can increase the proportion of bench-press concentric phase in which barbell is accelerated (AP) when combining FWR with EB. This is relevant since minimizing the deceleration portion of resistance training exercises seems to be a key component in transferring the training effects to competition performance. Although the benefits observed in the current research have been obtained from a specific combination of EB (~80%) and FWR (~20%), future research should test different combinations and different intensities, in order to reinforce the present data. Finally, training studies will be necessary to ascertain the extent and nature of transfer consequent to the respective training protocols.

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FIGURE LEGENDS

Figure 1. Band configuration used during the bench press exercise

Figure 2. Maximal percentage of the concentric phase in which barbell is accelerated during bench press sets in both conditions, for rugby players and control group. Values are means $\pm$ SD. * Significantly different from FWR condition ($p < 0.05$). ** Significantly different from FWR condition ($p < 0.01$). # Significantly different from increase in control group ($p < 0.05$)

Figure 3. Maximal velocity in both conditions, for rugby players and control group. Values are means $\pm$ SD. * Significantly different from FWR condition ($p < 0.05$)

Table 1. Subject characteristics. Values are means $\pm$ SD. * different from control group, $P < 0.01$. ** different from control group, $p < 0.001$

Group	Age (years)	Weight (Kg)	Height (cm)	Bench press 1RM (Kg)
Rugby players (N=8)	27 $\pm$ 4.15*	101.33 $\pm$ 12.13**	185.17 $\pm$ 3.54	142.33 $\pm$ 14.24**
Controls (N=8)	20.86 $\pm$ 1.35	74.71 $\pm$ 7.04	181.00 $\pm$ 4.90	78.29 $\pm$ 10.07

ACCEPTED

Table 2. Maximal repetitions, average velocity and average AP results. Values are means $\pm$ SD. * different from FWR condition, $p < 0.05$.

	Rugby players		Controls	
	FWR	EB+FWR	FWR	EB+FWR
Number of repetitions	6.1 $\pm$ 2.1	7 $\pm$ 1.67	5 $\pm$ 1.15	6.14 $\pm$ 2.12
Average velocity (m·s ⁻¹)	0.172 $\pm$ 0.03	0.197 $\pm$ 0.15 *	0.173 $\pm$ 0.02	0.185 $\pm$ 0.02
Average accelerative %	50.52 $\pm$ 6.61	54.46 $\pm$ 6.01	50.82 $\pm$ 4.68	54.67 $\pm$ 4.78

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